

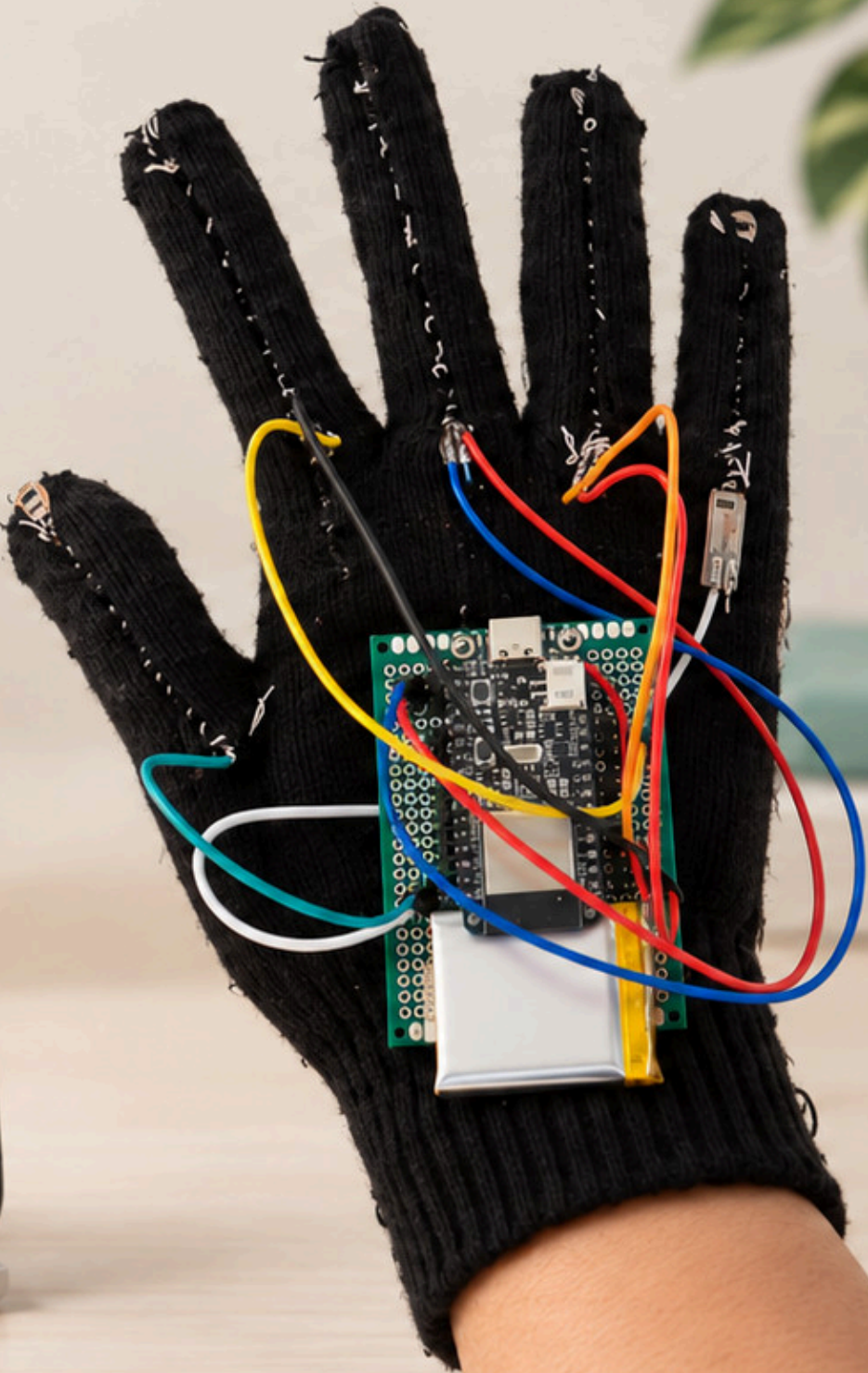
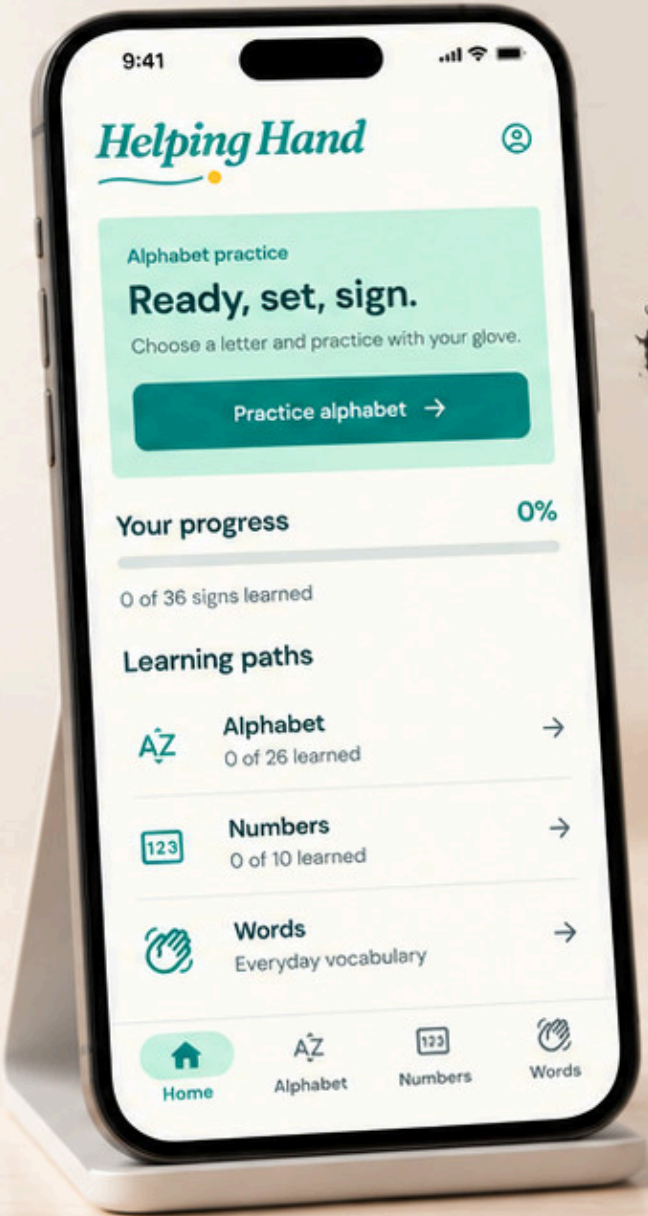
BETA BUILD · OCTOBER 6, 2026

Helping Hand

A Wearable ASL Learning System

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THE PROBLEM

Learning ASL needs feedback.

>90%

of deaf children are born to hearing parents.

Books and videos show the sign.

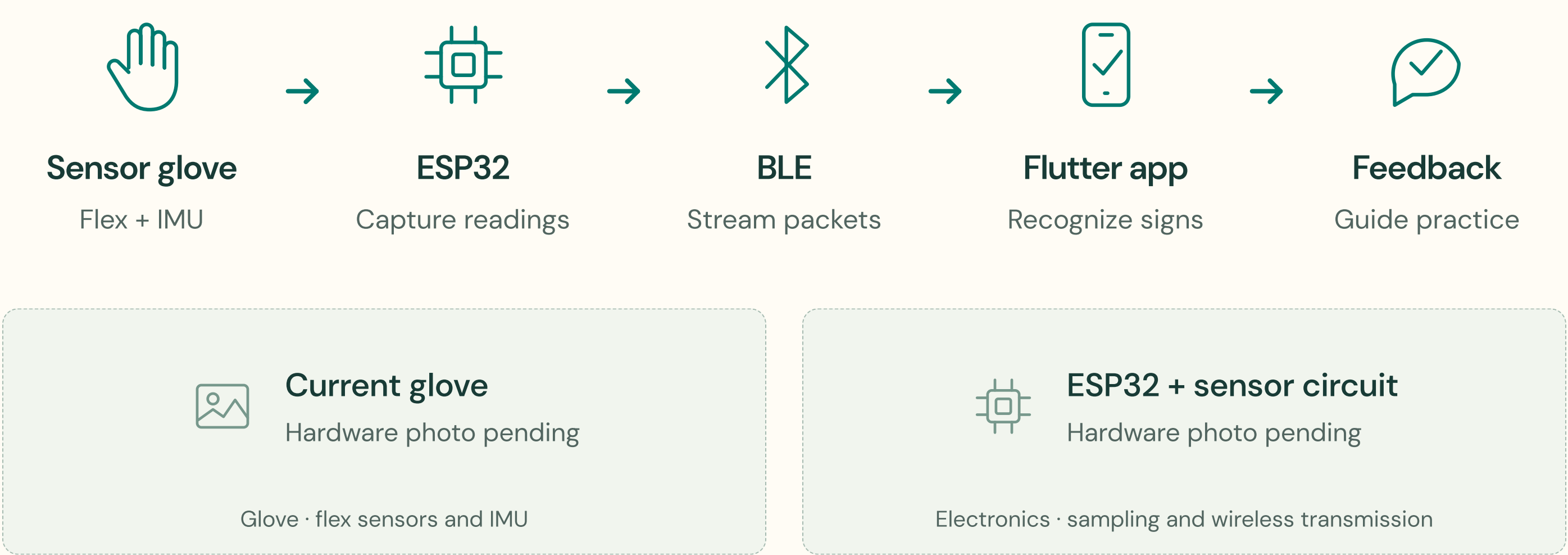
2D examples cannot assess your hand position.

No live correction from the reference alone.



OUR SOLUTION

Wear. Sign. Get feedback.



The first complete pipeline



Alpha glove



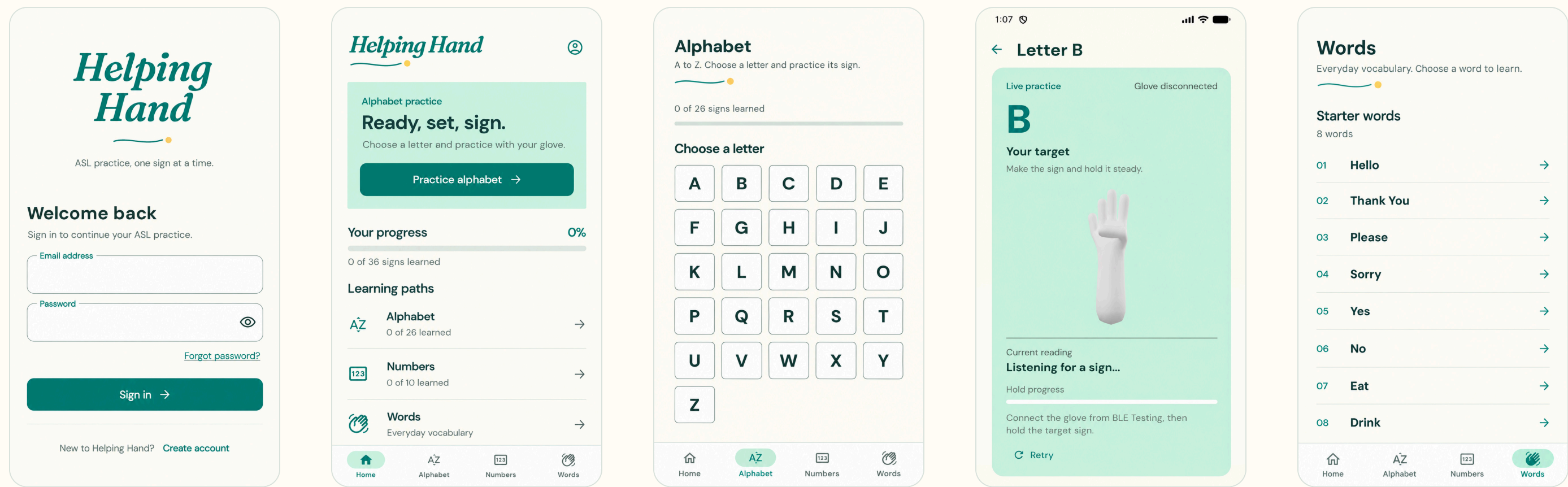
Alpha interface

- 01 **Sensor capture**
Finger bend + motion readings
- 02 **Wireless streaming**
Glove → BLE → app
- 03 **Static classification**
36-class TFLite baseline

THE BETA APP

A clearer path to practice

Firebase sign-in · dedicated practice pages · account-specific progress



01 Sign in

02 Choose a path

03 Pick a target

04 Practice

05 Explore words

Recognizing words with a sequence model

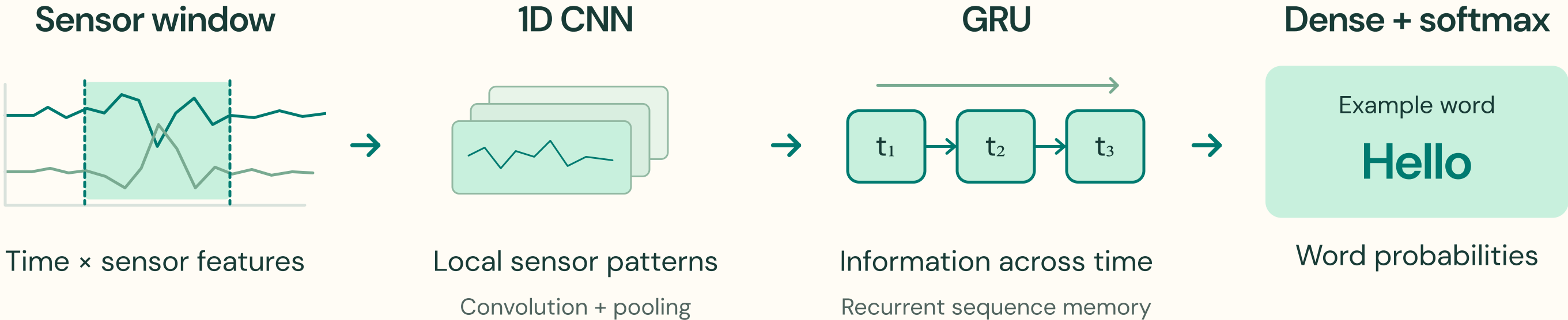
Alpha · MLP

One sensor snapshot → static class

Beta · sequence model

Sensor window → word prediction

Proposed CNN-GRU pipeline



CNN-LSTM · comparison candidate

Demo target · 3–4 beginner words

HELPING HAND IN ACTION

LIVE DEMO!!!

Static sign → Hello

Beta features and acceptance rules

Completed implementation

Core experience

- 36-class static baseline
- Firebase accounts + lessons
- Account-scoped static progress

In progress

Dynamic words

- 3–4-word demo target
- Sequence-model comparison
- Orientation + signer robustness

Planned

Tactile guidance

- 10 planned piezo elements
- Error-to-vibration mapping
- Calibration + user testing

Static completion rules

$\geq 80\%$

Confidence

$\geq 750\text{ ms}$

Stable matching hold

≥ 5

Matching packets

$\leq 250\text{ ms}$

Gap between packets

NEXT MILESTONE

Feel what needs correcting.

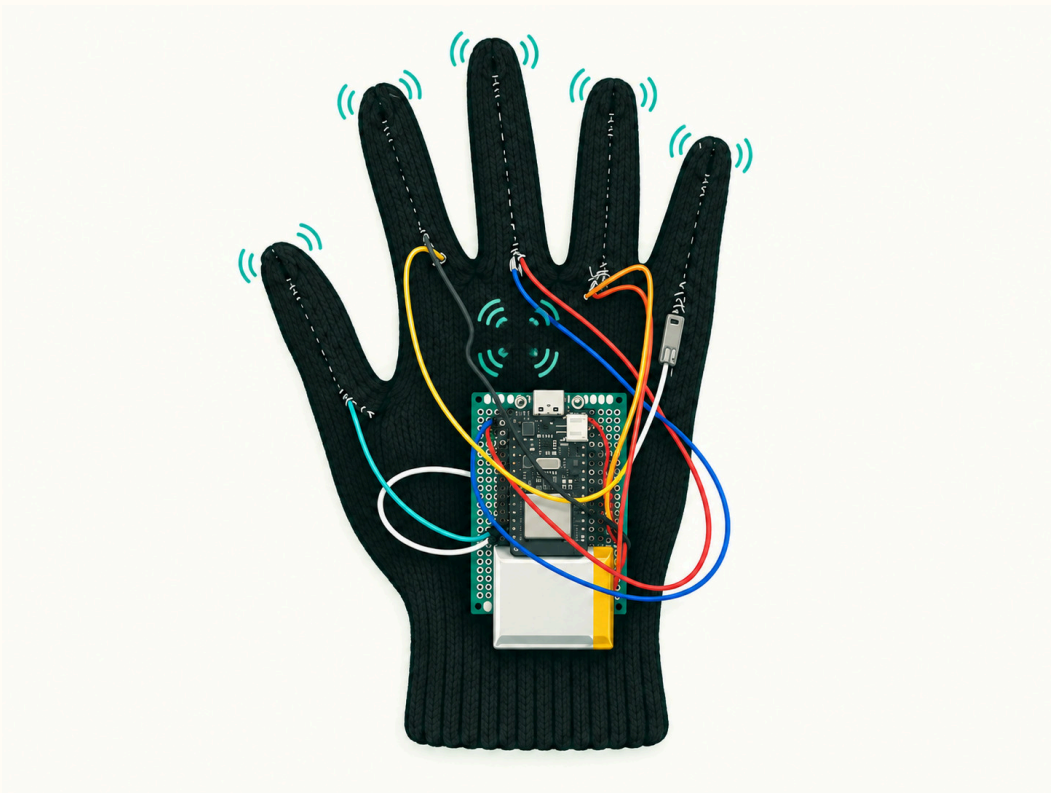
5 + 5

X pattern + one on each finger

Planned piezo feedback for incorrect positioning.

Estimated deviation → Pulse pattern / intensity

Placement, error mapping, and tactile response need calibration.



Concept illustration · planned feature

Oct 7–13	Oct 14–20	Oct 21–27	Oct 28–Nov 3
Mount + wire	Calibrate guidance	Refine recognition	Test + polish
Actuators and drivers	Error → tactile response	Orientation + signer trials	User trials and release prep

Helping Hand

Thank you. Questions?

Practice with feedback.

Next: guidance you can feel.